Yolov7 模型验证（训练+推理）

环境信息

<table>
<tr><th>服务器IP</th><td colspan="3">10.10.129.7</td></tr>
<tr><th rowspan="3">Hardware</th><td rowspan="2">Host</td><td>CPU</td><td>Intel(R) Xeon(R) Gold 6330 CPU @ 2.00GHz</td></tr>
<tr><td>Memory</td><td>500GB</td></tr>
<tr><td>Device</td><td>MLU</td><td>推理：MLU370-S4
训练：MLU370-X8</td></tr>
<tr><th rowspan="6">Software</th><td>OS</td><td colspan="2">Centos 7.6</td></tr>
<tr><td>Kernel</td><td colspan="2">3.10.0-957</td></tr>
<tr><td>Driver</td><td colspan="2">v5.10.10</td></tr>
<tr><td>docker-images:</td><td colspan="2">yellow.hub.cambricon.com/pytorch/pytorch:v1.14.0-torch1.9-ubuntu18.04-py37</td></tr>
<tr><td>Real Data</td><td colspan="2">nas</td></tr>
<tr><td>toolkit</td><td colspan="2">1.14</td></tr>
</table>

测试方法

- **代码模型数据集准备**

模块	链接	版本	备注
代码	https://github.com/WongKinYiu/yolov7.git	master	
模型	https://github.com/WongKinYiu/yolov7/releases/download/v0.1/yolov7-w6.pt	yolov7-w6.pt	yolov7-w6.pt
数据集	COCO2017	COCO2017	

代码适配

- **MLU**

```
git clone https://github.com/WongKinYiu/yolov7.git

python /torch/src/catch/tools/torch_gpu2mlu.py -i yolov7/
```

其它参见FAQ章节

验证测试（MLU）

安装依赖

```
#torch
pip install pycocotools
```

数据集

```
ln -sf /datasets/COCO2017/ coco
```

备注：如果使用NASCOCO2017 没有写入权限，可以先当前目录创建一个coco文件夹（文件夹有写权限），在将COCO2017 目录内容挂在到coco目录内即可。

推理

- **MLU370**

```
wget https://github.com/WongKinYiu/yolov7/releases/download/v0.1/yolov7-w6.pt

python test.py --data data/coco.yaml --img 640 --batch 32 --conf 0.001 --iou 0.65 --device 0 --weights yolov7-w6.pt --name yolov7_640_val
```

```
(pytorch) root@node-deploy2:/home/share/shoucai/yolov7_mlu# python test.py --data data/coco.yaml --img 640 --
batch 32 --conf 0.001 --iou 0.65 --device 0 --weights yolov7-w6.pt --name yolov7_640_val
[WARNING][/torch/catch/torch_mlu/csrc/aten/util/version.cpp:134][operator()][process:645][thread:
139992966248256]: Cambricon NEUWARE minimum version requirements not met! Require DRIVER minimum verion is
5.10.10-1, but current version is 5.10.6-1
/torch/venv3/pytorch/lib/python3.7/site-packages/torch/_utils.py:110: UserWarning:  MLU operators don't support
64-bit calculation. so the 64 bit data will be forcibly converted to 32-bit for calculation.  (Triggered
internally at  /torch/catch/torch_mlu/csrc/aten/util/tensor_util.cpp:153.)
  return new_type(self.size()).copy_(self, non_blocking)
Fusing layers...
[2023-5-26 8:39:36] [CNNL] [Warning]:[cnnlGetConvolutionForwardAlgorithm] is deprecated and will be removed in
the future release. See cnnlFindConvolutionForwardAlgorithm() API for replacement.
Model Summary: 343 layers, 70394300 parameters, 70394300 gradients, 89.9 GFLOPS
 Convert model to Traced-model...
/torch/venv3/pytorch/lib/python3.7/site-packages/torch/nn/modules/module.py:561: UserWarning: The .grad
attribute of a Tensor that is not a leaf Tensor is being accessed. Its .grad attribute won't be populated
during autograd.backward(). If you indeed want the gradient for a non-leaf Tensor, use .retain_grad() on the
non-leaf Tensor. If you access the non-leaf Tensor by mistake, make sure you access the leaf Tensor instead.
See github.com/pytorch/pytorch/pull/30531 for more information.
  if param.grad is not None:
/torch/venv3/pytorch/lib/python3.7/site-packages/torch/nn/functional.py:723: UserWarning: Named tensors and all
their associated APIs are an experimental feature and subject to change. Please do not use them for anything
important until they are released as stable. (Triggered internally at  /torch/pytorch/c10/core/TensorImpl.h:
1176.)
  return torch.max_pool2d(input, kernel_size, stride, padding, dilation, ceil_mode)
 traced_script_module saved!
 model is traced!

val: Scanning 'coco/val2017.cache' images and labels... 5000 found, 0 missing, 48 empty, 0 corrupted: 100%||
5000/5000 [00:00<?, ?it/s]
               Class      Images      Labels           P           R      mAP@.5  mAP@.5:.95:   0%
|
| 0/157 [00:00<?, ?it/s][2023-5-26 8:39:42] [CNNL] [Warning]:[cnnlSetNmsDescriptor_v2] is deprecated and will
be removed in the future release, please use [cnnlSetNmsDescriptor_v5] instead.
[2023-5-26 8:39:42] [CNNL] [Warning]:[cnnlClip] is deprecated and will be removed in the future release, please
use [cnnlClip_v2] instead.
               Class      Images      Labels           P           R      mAP@.5  mAP@.5:.95: 100%|| 157/157
[02:12<00:00,  1.18it/s]
                 all        5000       36780       0.724       0.622       0.671       0.485
Speed: 4.1/2.5/6.6 ms inference/NMS/total per 640x640 image at batch-size 32

Evaluating pycocotools mAP... saving runs/test/yolov7_640_val14/yolov7-w6_predictions.json...
loading annotations into memory...
Done (t=0.61s)
creating index...
index created!
Loading and preparing results...
DONE (t=4.55s)
creating index...
index created!
Running per image evaluation...
Evaluate annotation type *bbox*
DONE (t=68.79s).
Accumulating evaluation results...
DONE (t=13.13s).
 Average Precision  (AP) @[ IoU=0.50:0.95 | area=   all | maxDets=100 ] = 0.501
 Average Precision  (AP) @[ IoU=0.50      | area=   all | maxDets=100 ] = 0.680
 Average Precision  (AP) @[ IoU=0.75      | area=   all | maxDets=100 ] = 0.541
 Average Precision  (AP) @[ IoU=0.50:0.95 | area= small | maxDets=100 ] = 0.301
 Average Precision  (AP) @[ IoU=0.50:0.95 | area=medium | maxDets=100 ] = 0.553
 Average Precision  (AP) @[ IoU=0.50:0.95 | area= large | maxDets=100 ] = 0.693
 Average Recall     (AR) @[ IoU=0.50:0.95 | area=   all | maxDets=  1 ] = 0.379
 Average Recall     (AR) @[ IoU=0.50:0.95 | area=   all | maxDets= 10 ] = 0.619
 Average Recall     (AR) @[ IoU=0.50:0.95 | area=   all | maxDets=100 ] = 0.668
 Average Recall     (AR) @[ IoU=0.50:0.95 | area= small | maxDets=100 ] = 0.483
 Average Recall     (AR) @[ IoU=0.50:0.95 | area=medium | maxDets=100 ] = 0.732
 Average Recall     (AR) @[ IoU=0.50:0.95 | area= large | maxDets=100 ] = 0.843
Results saved to runs/test/yolov7_640_val14
```

训练

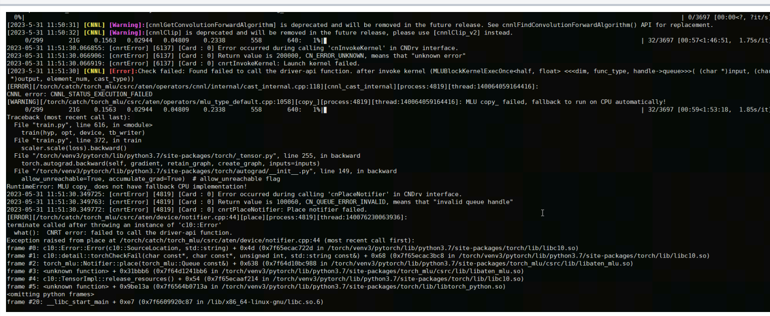
- **MLU370**

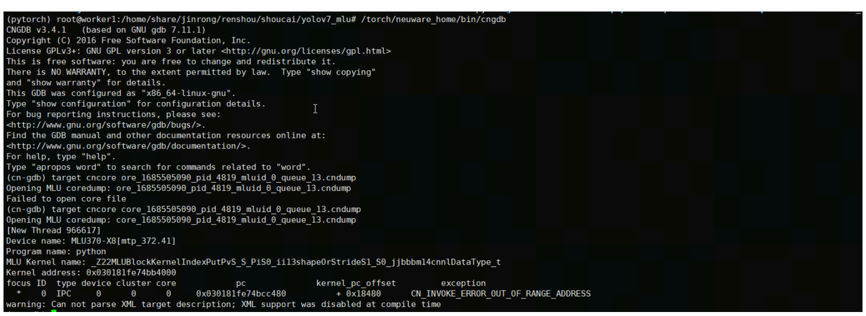
```
python  -m torch.distributed.launch --nproc_per_node=2 train.py --workers 8 --device 0,1 --batch-size 64 --data data/coco.yaml --img 640 640 --cfg cfg/deploy/yolov7.yaml --weights '' --name yolov7 --hyp data/hyp.scratch.p5.yaml --epochs 10
```

性能数据

模型	测试项	设备	设备数量	框架版本	运行方式	参数	输入规模	时延	精度	利用率（MLU）	内存（MLU）	备注
yolov7	推理	MLU370-X4	1	Pytorch 1.9	框架	--batch-size 32 --conf 0.001 --iou 0.65	640x640	Speed: 4.1/2.5/6.6 ms inference/NMS/total per 640x640	IoU=0.50:0.95 = 0.501 IoU=0.50 = 0.680	28%	6900 MiB/	
yolov7	训练	MLU370-X8	1	Pytorch 1.9	框架							
yolov7	训练	MLU370-X4	2	Pytorch 1.9	框架			01:29				

FAQ

<table>
<tr><th>ID</th><th>问题</th><th>复现方法</th><th>修改方法</th><th>原因</th></tr>
<tr><td>1</td><td>


</td>
<td>python train_aux.py --workers 16 --device 0 \
  --batch-size 16 --data data/coco.yaml \
  --img 1280 1280 --cfg cfg/training/yolov7-w6.yaml \
  --weights '' --name yolov7-w6 --hyp data/hyp.scratch.p6.yaml
X8 batch_size 16</td><td></td><td></td></tr>
</table>

2	/opt/conda/lib/python3.8/site-packages/torch/functional.py:598: UserWarning: torch.meshgrid: in an upcoming release, it will be required to pass the indexing argument. (Triggered internally at /opt/pytorch/pytorch/aten/src/ATen/native/TensorShape.cpp:2323.) return _VF.meshgrid(tensors, **kwargs) # type: ignore[attr-defined] Model Summary: 415 layers, 37622682 parameters, 37622682 gradients, 106.5 GFLOPS Scaled weight_decay = 0.0005 Optimizer groups: 95 .bias, 95 conv.weight, 98 other Traceback (most recent call last): File "train.py", line 616, in <module> train(hyp, opt, device, tb_writer) File "train.py", line 245, in train dataloader, dataset = create_dataloader(train_path, imgsz, batch_size, gs, opt, File "/home/share/yolov7/utils/datasets.py", line 69, in create_dataloader dataset = LoadImagesAndLabels(path, imgsz, batch_size, File "/home/share/yolov7/utils/datasets.py", line 392, in __init__ cache, exists = torch.load(cache_path), True # load File "/opt/conda/lib/python3.8/site-packages/torch/serialization.py", line 713, in load return _legacy_load(opened_file, map_location, pickle_module, **pickle_load_args) File "/opt/conda/lib/python3.8/site-packages/torch/serialization.py", line 920, in _legacy_load magic_number = pickle_module.load(f, **pickle_load_args) _pickle.UnpicklingError: STACK_GLOBAL requires str	训练/推理出现	删除数据集的.cache	